

OpenEBench4AI: Integrating Execution, Metrics, and Resource Flexibility for Benchmarking AI models



Vizcarra, Karla^{1,2}, Redondo, Anna^{1,2}, Fernández, José María^{1,2}, The OpenEBench Team, Gelpí, J.L.^{1,2,3}, Capella-Gutierrez S.^{1,2}

¹ Spanish National Bioinformatics Institute (INB/ELIXIR-ES).ES.

² Barcelona Supercomputing Center (BSC).ES.

³ Dept Biochemistry and Molecular Biomedicine, University of Barcelona (UB).ES.

OpenEBench4AI extends the **OpenEBench** platform with an **automated, scalable** framework for executing and benchmarking AI models. It supports diverse computing environments (CPU/GPU) and both public and private datasets, enabling **flexible, reproducible** evaluations.



Key New features

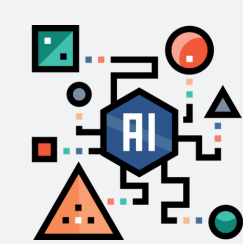
1. New modules for OpenEBench website
2. AI Benchmark Execution Platform
3. AI Submission Workflow Design
4. Integrated Metrics Implementation

Key features include a **user-friendly web** interface, **integrated metrics**, and a **streamlined experience** for participants and organizers. By combining execution, metrics, and resource management, OpenEBench4AI fosters a **transparent and efficient benchmarking ecosystem for AI research**.

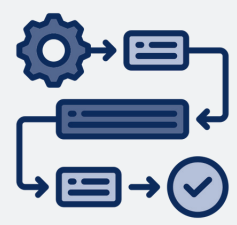
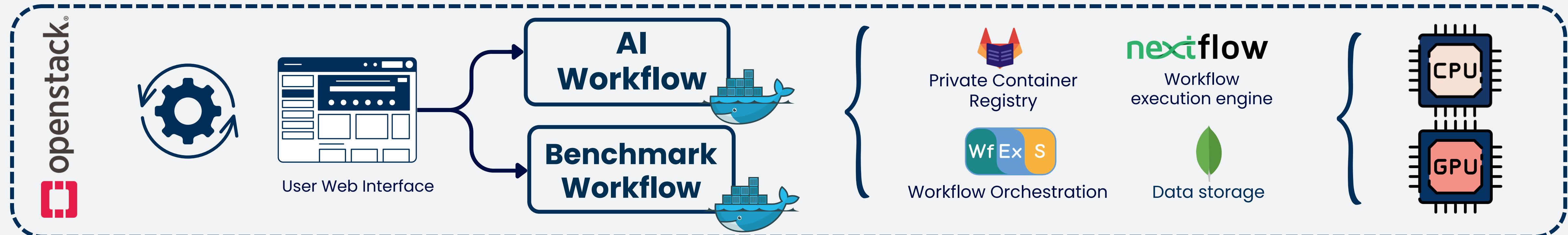


Web modules for AI

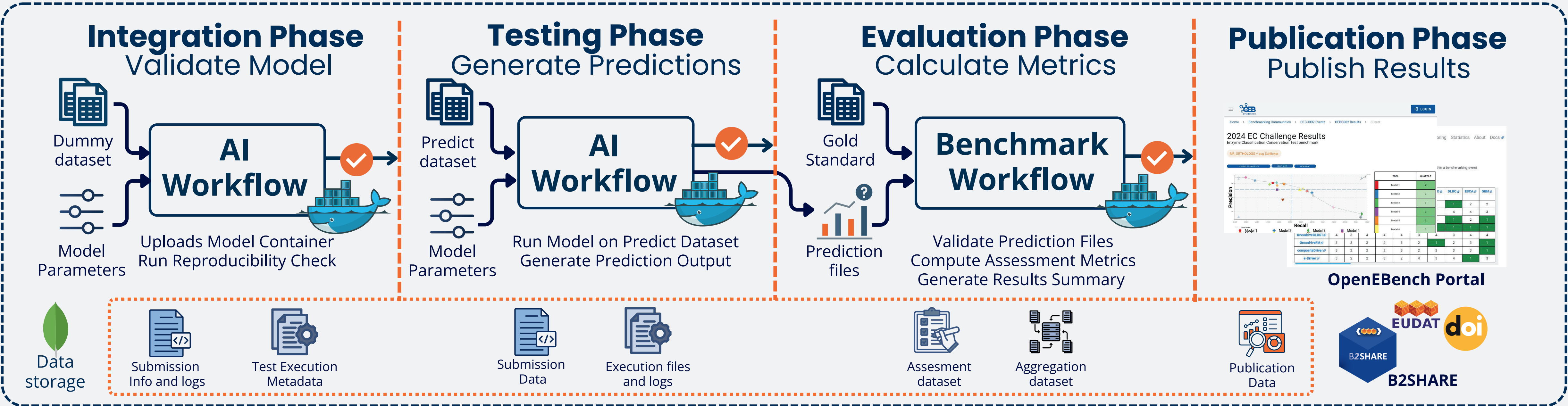
- ✓ Improved participant dashboard
- ✓ Participant enrollment for a challenge
- ✓ Participant submission
- ✓ Submission tracking status and configuration
- ✓ Challenge metrics configuration
- ✓ Workflow orchestration administration
- ✓ User administration
- ✓ AI platform administration and configuration



AI Benchmarking Execution platform



AI submission Workflow design



4



Integrated metrics implementation

- Metrics created and predefined following **FAIR principles**
- ✓ **Accesible** for selection during challenge configuration
 - ✓ **Automatically configured in the benchmark workflow**
 - ✓ **Ready** to be calculated without code manipulation
 - ✓ **Integrated** with web widgets at publication stage

Conclusions

- ✓ **OpenEBench4AI** enables **reproducible**, cross-platform **benchmarking of AI models** via a fully automated, container-based workflow.
- ✓ Simplifies challenge participation and **transparent performance** evaluation across diverse infrastructures.
- ✓ **Standardized metrics** and **robust execution** pipelines for FAIRer comparisons and more **trustworthy AI**.

Next steps

- ✓ Expand support for **multimodal** datasets and challenges that require **additional features**.
- ✓ Explore new AI cases: **Generative** model-based submissions, **federated** and **encrypted** data models.
- ✓ Enable automated integration of **custom metrics** for specific domains.

References

OpenEBench new web design



OpenEBench Widgets



WfExS



AI Workflow



Contact

Karla Vizcarra Loayza
karla.vizcarra@bsc.es
INB/ELIXIR-ES Coordination Hub
Barcelona Supercomputing Center (BSC)
OpenEBench Team
<https://openebench.bsc.es/team>



This initiative is part of the talent attraction and retention programs within investment 4 of component 19 of the Recovery, Transformation, and Resilience Plan, C005/24-ED CV1

